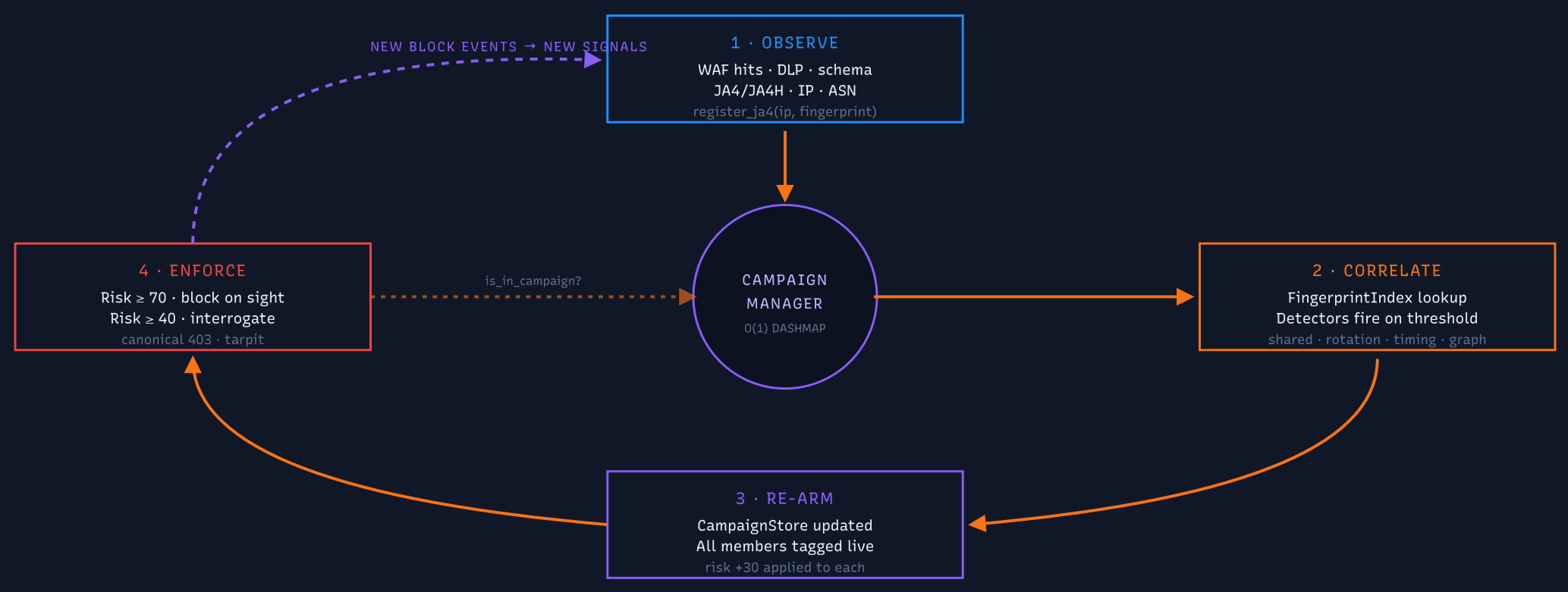
SYNAPSE EDGE DEFENSE

Threat Intel Feedback Loop

How signals observed at one IP turn into blocks at every related IP. A closed loop from detection to correlation to re-armed enforcement, running in-process at the edge — O(1) fingerprint lookups, no backend round-trip.

8	4	O(1)	+30
CORRELATION DETECTORS	SIGNAL CATEGORIES	FINGERPRINT LOOKUP	CAMPAIGN RISK ADD

THE CLOSED LOOP — OBSERVE · CORRELATE · RE-ARM · ENFORCE



SIGNAL CATEGORIES — WHAT FEEDS THE LOOP

ATTACK
WAF blocks, injection attempts, known-exploit matches

ANOMALY
Schema drift, unusual headers, TLS/UA mismatches

BEHAVIOR
Rate violations, session anomalies, auth patterns

INTELLIGENCE
Threat feeds, Tor exits, datacenter ranges, bad-IP lists

CORRELATION DETECTORS — 8 RUNNING IN-PROCESS

shared_fingerprint
JA4+JA4H COLLISION
3+ IPs sharing the same combined fingerprint → campaign. Default threshold: 3.

ja4_rotation
ACTOR FLIP PATTERN
One IP cycling through 3+ JA4s fast → evasion attempt.

timing_correlation
SIMULTANEOUS ACTIONS
Requests arriving within tight windows across IPs suggest scripted coordination.

network_proximity
ASN / SUBNET
Actors clustered in the same /24 or ASN get linked even without fingerprint overlap.

behavioral_similarity
REQUEST SHAPE
Matching path sequences, body shapes, and header orderings across IPs.

attack_sequence
KILL-CHAIN PATTERN
Same ordered attack progression seen across multiple actors.

auth_token
SHARED CREDENTIALS
Multiple IPs presenting the same session/JWT → stuffing or account takeover.

graph
RELATIONSHIP MESH
Transitive links — if A↔B and B↔C, all three enter the same campaign.

BEFORE THE LOOP VS AFTER

WITHOUT FEEDBACK

198.51.100.12 blocked

Original attacker IP gets a 403. The other 17 IPs sharing its JA4 + JA4H fingerprint continue unblocked until they individually trip a WAF rule or the per-IP risk score climbs — potentially minutes of exposure per actor.

→

WITH FEEDBACK

17 IPs blocked on sight

Campaign detection fires on the 3rd IP to share the fingerprint. Every member of the set — past, present, and any future IP that joins — gets the campaign membership tag. Their next request sees risk 70+ without any per-IP learning period. The loop closes at edge speed, no backend consultation required.

Detect once. Block everywhere. At edge speed.
Eight detectors, four signal categories, one in-process fingerprint index. No round-trip to a backend — the loop closes before the next request arrives.